Homework 5

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March 2026

Homework code located at: https://github.com/nguyencquan/Markov2026

1 Eigenvalues of Stochastic Matrices

a) Show that p has at least one eigenvalue equal to 1.
If p is a stochasitc matrix size NxN, then all of the rows sum up to 1. We are trying to find some matrix v such that $pv = v$. Expanding this matrix multiplication we get a specific index of v being

$$v_j = \sum_{i=1}^{N} p_{j,i} v_i$$

However we know that $\sum_{i=1}^{n} p_{j,i} = 1, \forall j$, hence if we define every vector in v as 1, then we have

$$v_j = \sum_{i=1}^{N} p_{j,i} = 1$$

Therefore, for any stochastic matrix, an eigenvalue of 1 will have a corresponding matrix with a column length N being composed of 1, as all rows in the stochastic matrix sum to 1.
b) Show that all the other eigenvalues have magnitudes less then or equal to 1.
The Gershgorin circle theorem states that, if $R_i = \sum_{i \neq j} |a_{ij}|$, where i is some row of the matrix A, then all eigenvalues of A exists in at least one gershgorin disk denoted by $(D(a_{ii}, R_i))$
Looking at a stochastic matrix which has all positive entries and all rows sum up to one, we know for all i

$$1 = \sum_{j=1}^{N} a_{ij} = a_{ii} + \sum_{i \neq j} a_{ij} = a_{ii} + \sum_{i \neq j} |a_{ij}|$$

Hence the maximum upper bound for this Gershgorin disk is 1, and is always 1.
Furthermore, since $0 \leq a_{ii} \leq 1$ we can calculate the lower bound from

$$\begin{aligned} &1 - a_{ii} = R_i \\ &\implies 0 \leq R_i \leq 1 \\ &\implies 1 \geq a_{ii} - R_i \geq -1 \end{aligned}$$

From the first part we see the largest possible upper bound is 1 and lowest lower bound is -1 hence the absolute value of all eigenvalues must be equal to or less then 1.
c) Show that if p1 and p2 are stochastic matrices of the same size, p1p2 is also stochastic

In order to be a stochastic matrix, the sum of rows must equal 1 hence if we look at a single row.

$$\begin{aligned}
1 = \sum_{j=1}^{\infty} p1p2_{i,j} &= \sum_{k=1}^{N}\sum_{j=1}^{N} p1_{i,j}p2_{j,k} \\
&= \sum_{j=1}^{N}\sum_{k=1}^{N} p1_{i,j}p2_{j,k} \\
&= \sum_{j=1}^{N} p1_{i,j} \sum_{k=1}^{N} p1_{j,k} \\
&= \sum_{j=1}^{N} p1_{i,j} = 1
\end{aligned}$$

However this applies to all rows so the product of these two matrices are stochastic.

2 Three versions of the stationary distribution

a) Used detailed balance to find the stationary distribution
I am going on a limb and saying that you can have detailed balancing for this markov chain. By looking and the first and second node, we can setup the first detailed balancing focussed around 1 or specifically. Note the loop to self is ignored since probability leaving and entering cancels out.

$$\pi_1 p_1 = \pi_2 q_2 \implies \pi_2 = \pi_1 \frac{p_1}{q_2} = \pi_1 \frac{e^a}{e^b}$$

and furthermore

$$\pi_2 p_2 = \pi_3 q_3 \implies \pi_3 = \pi_2 \frac{p_2}{q_3} = \pi_1 \frac{e^{3a}}{e^{3b}}$$

$$\pi_3 p_3 = \pi_4 q_4 \implies \pi_4 = \pi_3 \frac{p_3}{q_4} = \pi_1 \frac{e^{6a}}{e^{6b}}$$

$$\pi_4 p_4 = \pi_5 q_5 \implies \pi_5 = \pi_4 \frac{p_4}{q_5} = \pi_1 \frac{e^{10a}}{e^{10b}}$$

So we have the following equation, note the exponential is a serial sum.

$$\pi(i) = \pi_1 e^{(a-b)\sum_{k=1}^{i-1} k}$$

However this needs to be normalized by

$$\frac{1}{1 + e^{(a-b)} + e^{3(a-b)} + e^{6(a-b)} + e^{10(a-b)}} = \frac{1}{e^{(a-b)\sum_{i=1}^{5} \frac{i(i-1)}{2}}}$$

Resulting in

$$\pi(i) = \frac{e^{(a-b)\frac{i(i-1)}{2}}}{\sum_{i=1}^{5} e^{(a-b)\frac{i(i-1)}{2}}}$$

b) Find the stationary distribution by numerically finding the eigenvalue of p^T corresponding to the eigenvalue 1, using $a = .04$, $b = .16$ and $K = .1$
So we are trying to solve for the eigenvector of p^T hence we want

$$\vec{\pi} p = \vec{\pi}$$

$$p^T \vec{\pi}^T = \vec{\pi}^T$$

$$(p^T - I)\vec{\pi}^T = 0$$

Solving this numerically in R, we get the eigenvector:

$$\vec{\pi} = [0.297, 0.263, 0.207, 0.144, 0.089]$$

c) Simulate the markov chain for at least 10^6 steps. Plot a histogram of the fraction of time that the markov chain spent in different states Running the chain 10^6 times and starting at 1, we get a distribution of

$$[0.298, 0.262, 0.207, 0.145, 0.0888]$$

This is extremely close to the theoreitcal value we calculated using a numerical method.

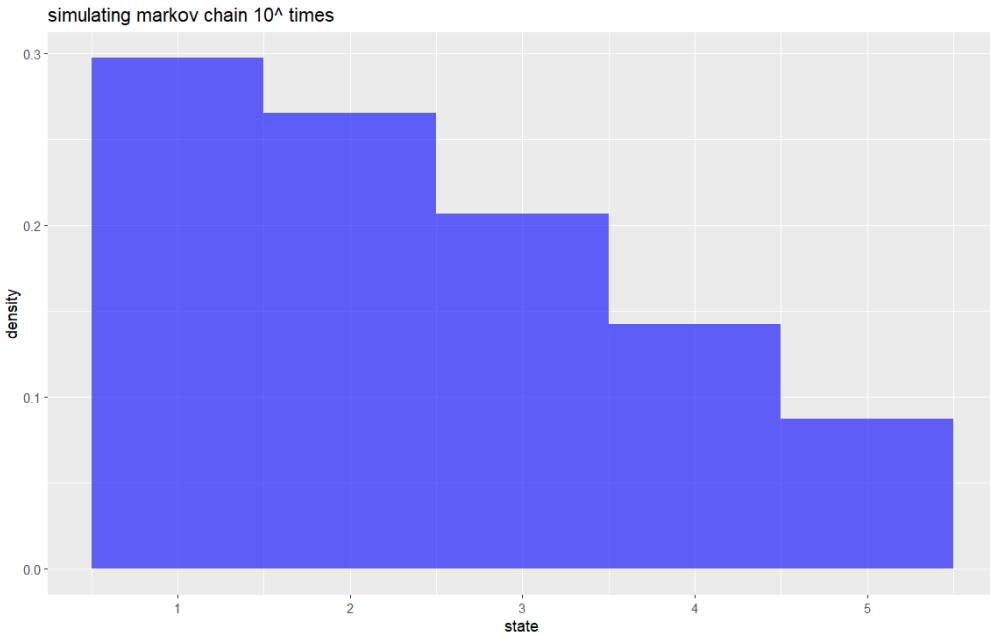

Figure 1: Simulation of markov chain

d) Plot your histogram from (c) together with the theoretical distributions found in (a) and (b).

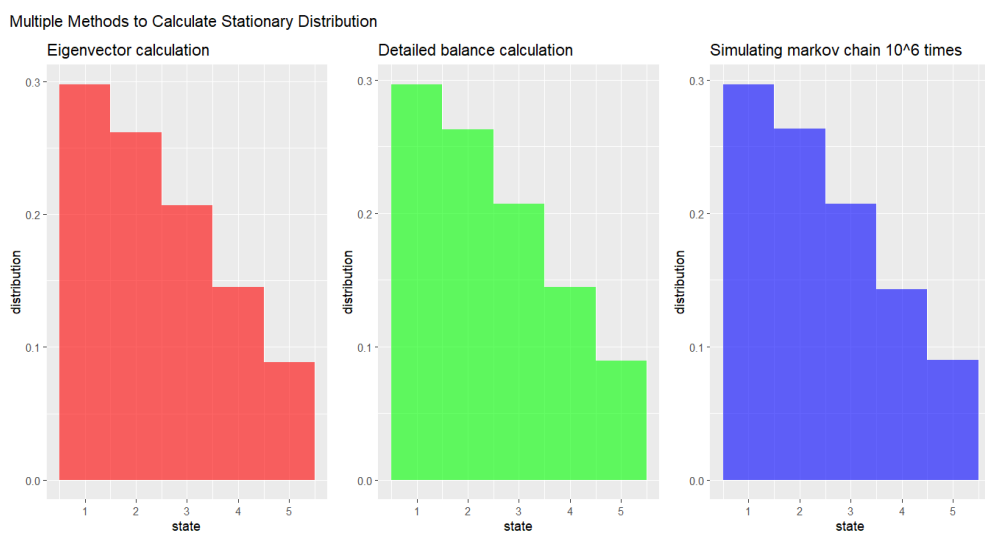

Figure 2: Simulation of markov chain

3 Convergence to the stationary distribution

a) Find the stationary distribution π
By utilizing detailed balance we have the following

$$\pi_1 a = \pi_2 a \implies \pi_1 = \pi_2 \tag{1}$$
$$\pi_2(1-a) = \pi_3(1-a) \implies \pi_2 = \pi_3 \tag{2}$$

Hence after normalization the stationary distribution is

$$\pi = \left[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right]$$

b) Find the eigenvectors and eigenvalues of the transpose of the probability transition matrix
We are given a probability transition matrix of

$$\begin{bmatrix} 1-a & a & 0 \\ a & 0 & 1-a \\ 0 & 1-a & a \end{bmatrix}$$

The transpose p^T is actually identical to the original matrix p. Hence solving for $p^T\vec{v} = \lambda\vec{v}$ through wolframalpha.com:

$$\begin{cases} \lambda_1 = 1 & \vec{v_1} = [1,1,1]^T \\ \lambda_2 = \sqrt{3a^2-3a+1} & \vec{v_2} = \left[-\frac{-1+2a-\sqrt{1-3a+3a^2}}{-1+a}, -\frac{-a+\sqrt{1-3a+3a^2}}{-1+a}, 1\right]^T \\ \lambda_3 = -\sqrt{3a^2-3a+1} & \vec{v_3} = \left[-\frac{-1+2a+\sqrt{1-3a+3a^2}}{-1+a}, -\frac{-a-\sqrt{1-3a+3a^2}}{-1+a}, 1\right]^T \end{cases}$$

c) show that q_n converges exponentially to π
Decomposing a starting distribution of everything being at 1 into eigenvectors, note I will be using R however the values will be adjusted to match that of wolfram alpha...

$$q_0^T = .3333v_1 + -0.3359v_2 + 0.0025v_3$$

However to make it easier to look at we will call this

$$q_0^t = \hat{v_1} + \hat{v_2} + \hat{v_3}$$

Now if we solve the next iteration:

$$q_1^T = A(\hat{v_1} + \hat{v_2} + \hat{v_3}) = A\hat{v_1} + A\hat{v_2} + A\hat{v_3} = \hat{v_1} + \lambda_2\hat{v_2} + \lambda_3\hat{v_3}$$
$$q_2^T = A^2(\hat{v_1} + \hat{v_2} + \hat{v_3}) = \hat{v_1} + \lambda_2^2\hat{v_2} + \lambda_3^2\hat{v_3}$$

Hence q_n must be

$$\begin{aligned} q_n^T &= .3333v_1 + \lambda_2^n\hat{v_2} + \lambda_3^n\hat{v_3} \\ &= \pi^T + \lambda_2^n\hat{v_2} + \lambda_3^n\hat{v_3} \end{aligned}$$

As with induction

$$\begin{aligned} q_{n+1}^T = Aq_n^T &= A(\pi^T + \lambda_2^n\hat{v_2} + \lambda_3^n\hat{v_3}) \\ &= A\pi^T + \lambda_2^n A\hat{v_2} + \lambda_3^n A\hat{v_3} \\ &= \pi^T + \lambda_2^{n+1}\hat{v_2} + \lambda_3^{n+1}\hat{v_3} \end{aligned}$$

However we know that as $n \to \infty$ and $|\lambda_2|, |\lambda_3| < 1$ so let $K = max\{|\lambda_2|, |\lambda_3|\}$

$$q_n^T - \pi^T = \lambda_2^n\hat{v_2} + \lambda_3^n\hat{v_3}$$
$$||q_n^T - \pi^T|| = ||\lambda_2^n\hat{v_2} + \lambda_3^n\hat{v_3}|| \leq ||\lambda_2^n\hat{v_2}|| + ||\lambda_3^n\hat{v_3}|| = |\lambda_2|^n||\hat{v_2}|| + |\lambda_3|^n||\hat{v_3}|| \leq K^n[||\hat{v_2}|| + ||\hat{v_3}||] \leq CK^n$$

As a result the distribution must decay at least as fast as an exponential to the stationary distribution.

d) For what value a is the convergence to the stationary distribution faster
Note the smaller $|\lambda|$ is, as long as it is not 1 (and there is only one 1), the faster the decay and we also know that

$$|\lambda_2| = |\lambda_3| = \sqrt{3a^2 - 3a + 1}$$

Take the derivatives of the interior function

$$\frac{df}{da} = 6a - 3$$
$$\frac{d^2 f}{da^2} = 6$$

This indicates that at .5, it is concave up, hence the minimum value is at .5.
In other words, the value for a with the fastest decay rate is .5 since this minimizes the absolute value of the eigenvalues.
e) Run the markov chain a bunch of times starting at state 1

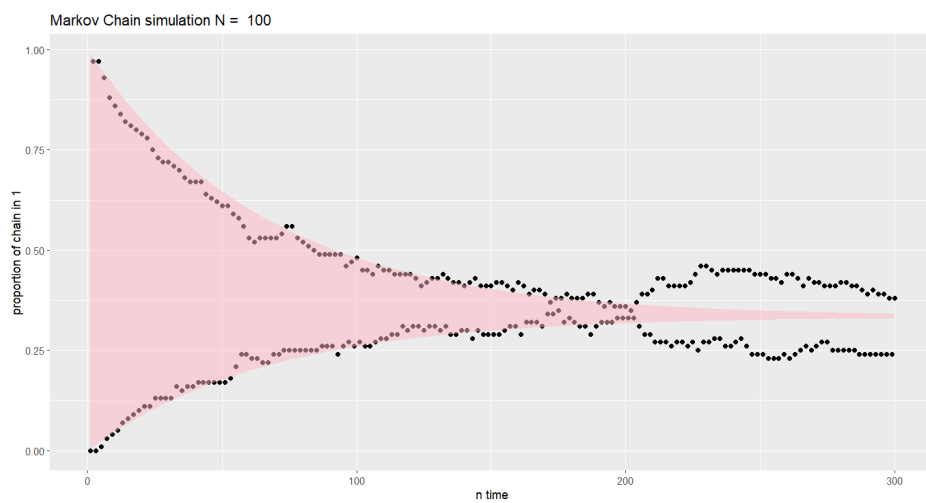

Figure 3: Markov chain simulation with $p = A$ and N = 100, the shaded region is the theoretical distribution

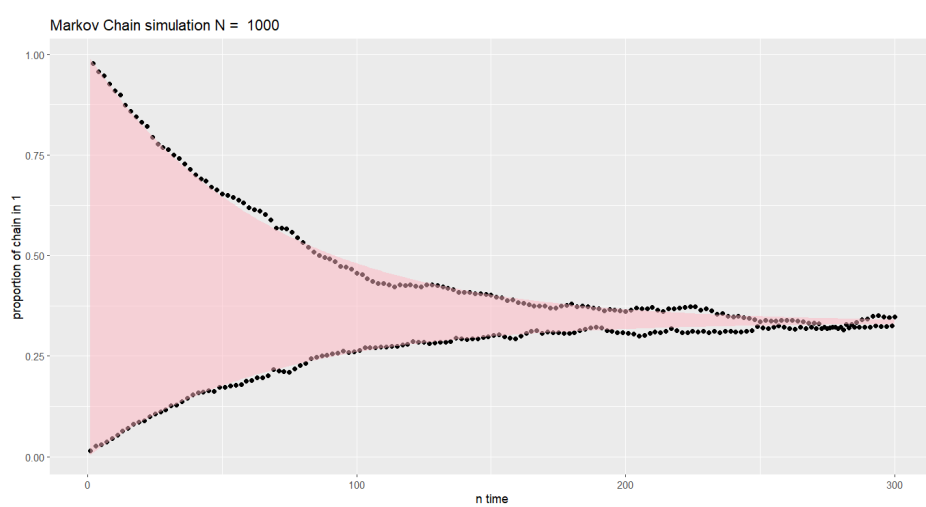

Figure 4: Markov chain simulation with $p = A$ and N = 1000, the shaded region is the theoretical distribution

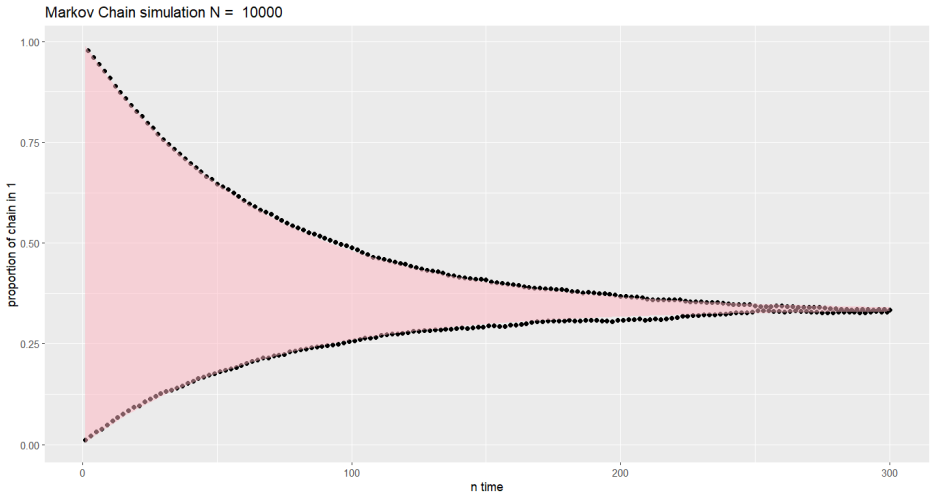

Figure 5: Markov chain simulation with $p = A$ and N = 10000, the shaded region is the theoretical distribution

4 Proof of Detailed balance for birth-death chains

Given the stationary distribution, we end up with the equation:

$$\pi P = [\pi_0(1-p_0-q_0)+\pi_1 q_1, \pi_0 p_0 + \pi_1(1-p_1-q_1)+\pi_2 q_2, ..., \pi_{N-1}p_{N-1}\pi_N(1-p_N-q_N)] = \pi$$

In other words, besides the boundary states we have

$$\pi_i = \sum_{k=0}^{N} \pi_k p(k,i) = \pi_{i-1}p_{i-1} + \pi_i(1-p_i-q_i) + \pi_{i+1}q_{i+1}$$

So starting with the intial case we have

$$P_0 = \pi_0(1-p_0-q_0) + \pi_1 q_1$$

Assume the n'th case is true such that

$$P_n = P_n + q_{n+1}\pi_{n+1} - p_n\pi_n$$

Then we have

$$\begin{aligned}
P_{n+1} &= \sum_{j=0}^{n+1} \pi_j \\
&= \pi_{n+1} + \sum_{j=0}^{n} \pi_j \\
&= \pi_{n+1} + P_n \\
&= \pi_{n+1}(1-p_{n+1}-q_{n+1}) + \pi_{n+2}q_{n+2} + \pi_n p_n + P_n \\
&= \pi_{n+1}(1-p_{n+1}-q_{n+1}) + \pi_{n+2}q_{n+2} + \pi_n p_n + P_n + q_{n+1}\pi_{n+1} - p_n\pi_n \\
&= \pi_{n+1} - \pi_{n+1}p_{n+1} + \pi_{n+2}q_{n+2} + P_n \\
&= (P_n + \pi_{n+1}) - \pi_{n+1}p_{n+1} + \pi_{n+2}q_{n+2} \\
&= P_{n+1} + \pi_{n+2}q_{n+2} - \pi_{n+1}p_{n+1}
\end{aligned}$$

hence by induction,

$$P_i = P_i + q_{i+1}\pi_{i+1} - p_i\pi_i$$